

Squares

Going over the calculations proposed in my earlier document it occurred to me that I hadn't looked in a great deal of depth at the squares used to help move from plain text to the encoding character so I have been visiting this today.

There is an interesting thing about these squares, they split the alphabet into 4 with two outliers and every letter in the partial alphabet can be reached from every other letter in the partial alphabet.

Looking for example at A, if we square A we get:

Y	A
A	C

Seems innocuous enough but what happens if we square Y?

A	Y
Y	W

Now A has moved to the top left corner and W has appeared.

Following this chain on to completion we get:

Y	A		A	Y		C	W
A	C		Y	W		W	Q
W	C		I	Q		Q	I
C	I		Q	Y		I	A

After this it starts looping back on it's self effectively exhausting this alphabet.

If we follow this procedure for the rest of the alphabet we end up with the following set of alphabets.

1	A Y C W Q I
2	B X F T H R
3	V D L N J P
4	U E O K S G

These alphabets seem to hold true regardless of the position but for those who haven't read my earlier PDF the math is given as:

Top Right:

$$x_1 = (p - a) \bmod 26$$

$$y_1 = (((26 - x) + a) + 1) \bmod 26$$

Bottom Left:

$$x_2 = (p + a) \bmod 26$$

$$y_2 = (((26 - x) + a) + 1) \bmod 26$$

$$\text{Top Left} = (x_1, y_2)$$

$$\text{Bottom Right} = (x_2, y_1)$$

Within these subsets, M and Z are unique in that they both square to themselves. This means that they do not fit into any of the 4 supplied alphabets. This leaves 24 letters in the primary alphabet so let's look at how that maps to the subset alphabets. For the purposes of illustration I've left M and Z in place.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	1	3	4	2	4	2	1	3	4	3	^	3	4	3	1	2	4	2	4	3	1	2	1	^

Notice what happens to the sets either side of the M? The numbers underneath have mirrored each other.

Sliding N – Z under A – M we now get the following table:

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	1	3	4	2	4	2	1	3	4	3	^
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
3	4	3	1	2	4	2	4	3	1	2	1	^

I haven't worked out exactly where this is going yet but looking for a way to give provide the extrapolate B from A, perhaps this will aid in a solution.